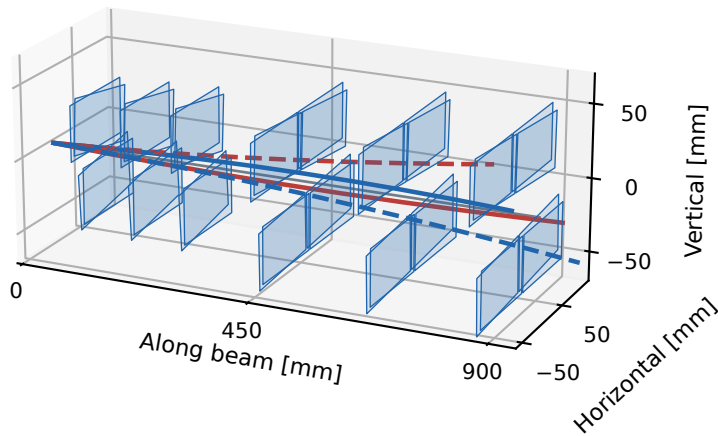


HPS A' acceptance: geometry, magnetic transport and selected spectra | v5.7.1

A | Active faces and passing e^+ (red) / e^- (blue): solid low-mass, dashed high-mass examples

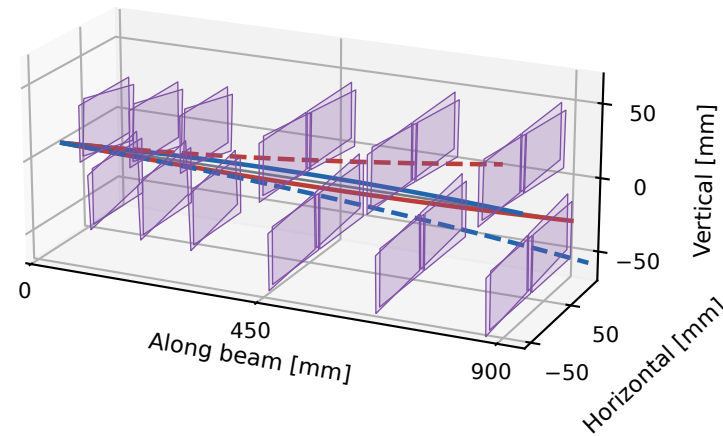
HPS 2015 | 1.056 GeV



Passing examples: 20 and 195 MeV

Nominal 15 mrad scale ≈ 15.84 MeV

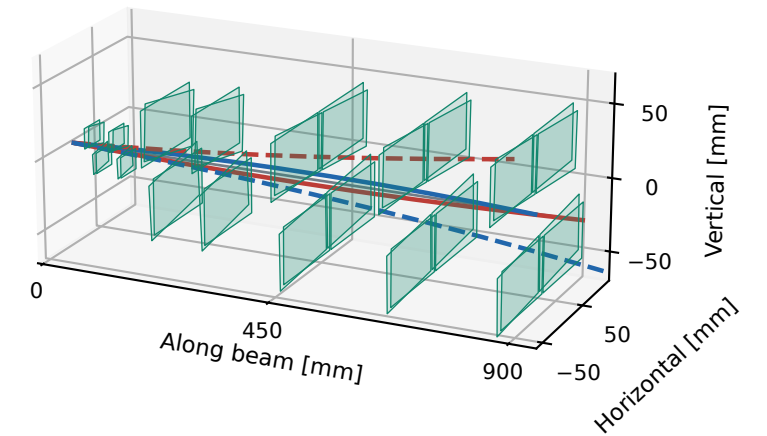
HPS 2016 | 2.300 GeV



Passing examples: 40 and 425 MeV

Nominal 15 mrad scale ≈ 34.50 MeV

HPS 2021 | 3.740 GeV

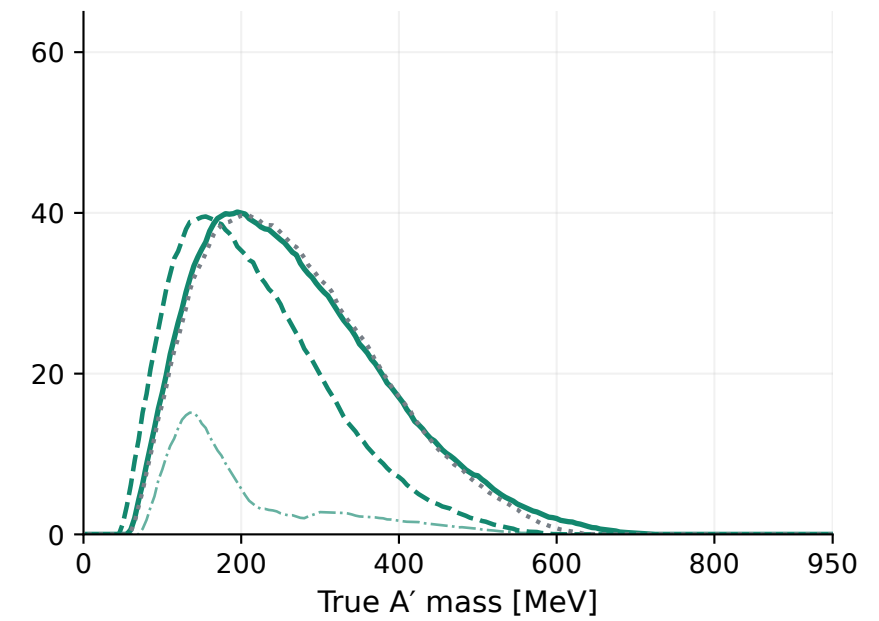
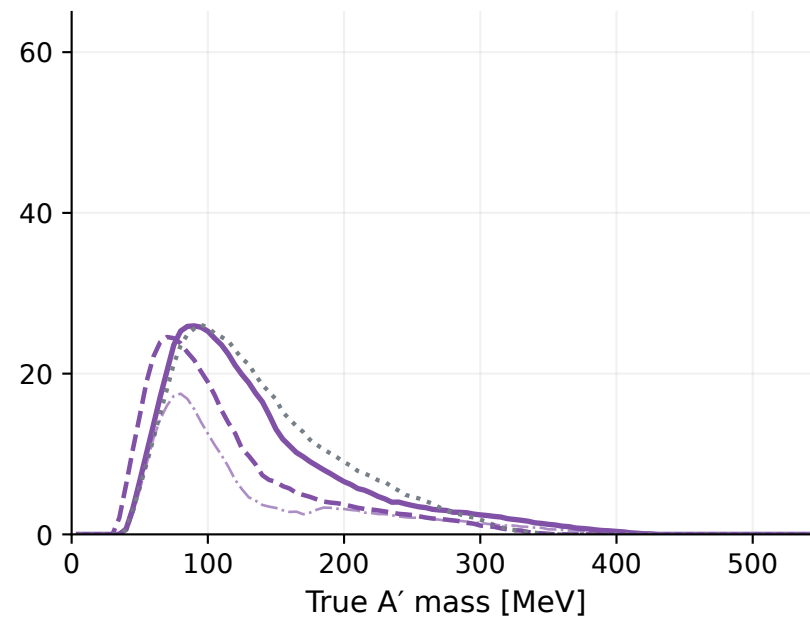
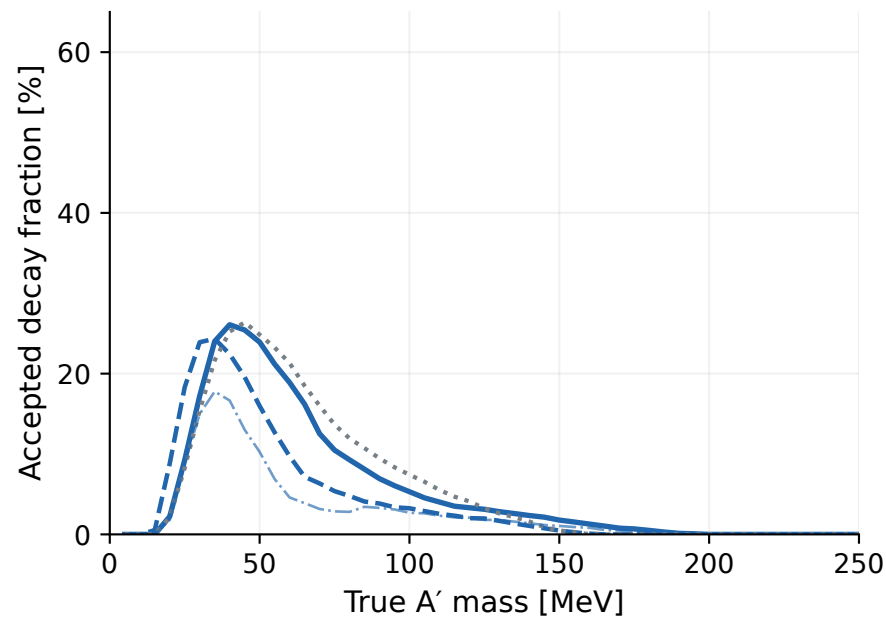


Passing examples: 60 and 715 MeV

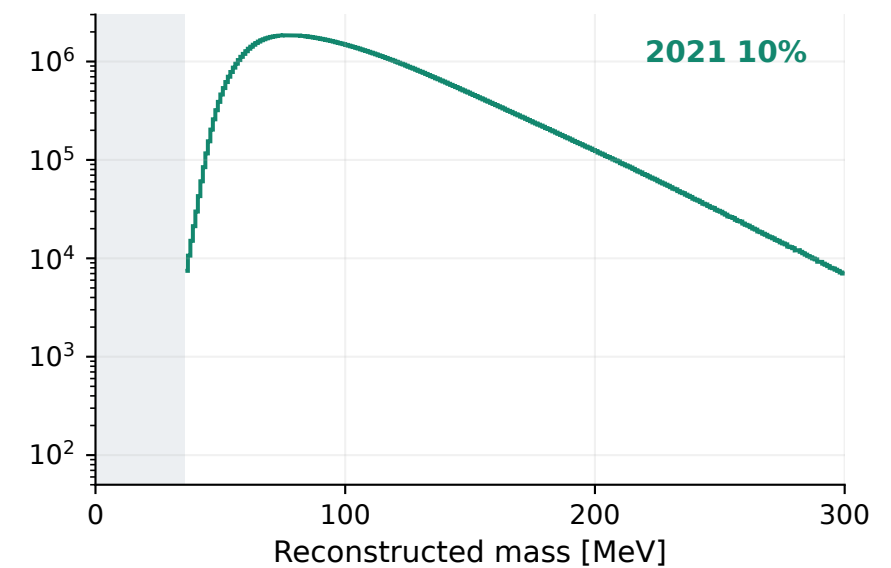
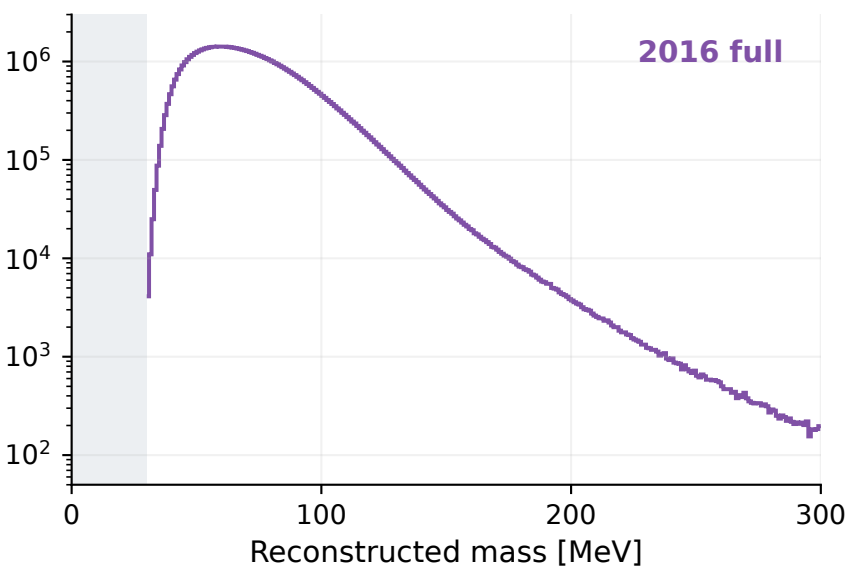
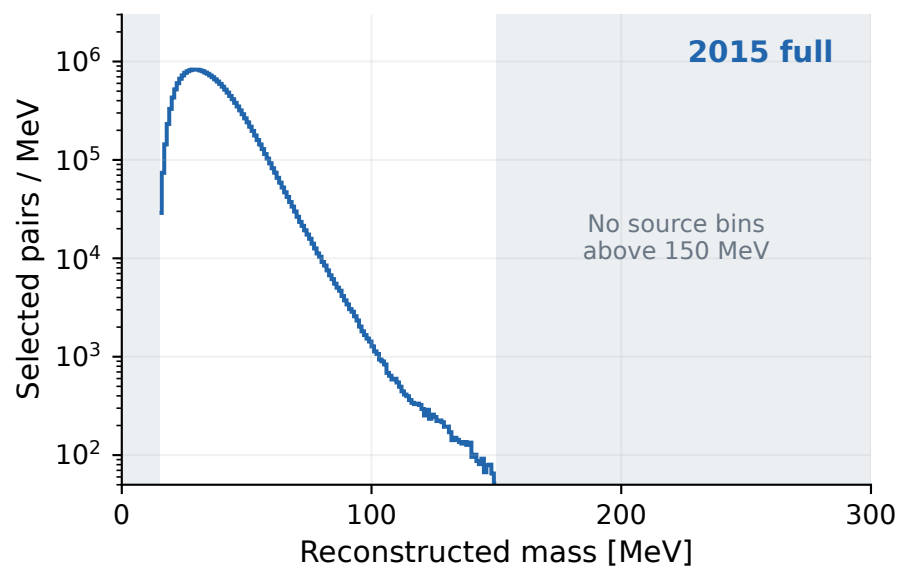
Nominal 15 mrad scale ≈ 56.10 MeV

B | Decay fraction through the magnetic field and required sensor views

— Field + hit cuts, $x = 1$ - - - Field + hit cuts, $x = 0.8$ B = 0 + hit cuts, $x = 1$ - · - · Field + all stations, $x = 1$



C | Existing selected-pair spectra



Hit cuts: 2015 ≥ 5 paired stations each + e^+ L1/L2; 2016 ≥ 5 each; 2021 ≥ 10 e^+ / ≥ 8 e^- individual strip hits.

Illustrated low/high masses are sampled passing examples. Curves include the field; material, hit finding, trigger and reconstruction remain omitted.